CENG 463 - Introduction to Machine Learning

Homework 3: CIFAR-10 Image Clustering

1. Introduction

This report presents the analysis and results of unsupervised learning techniques applied to the CIFAR-10 dataset. The objective of this homework is to explore clustering algorithms on image data, perform dimensionality reduction, and visualize the results to understand how different algorithms group similar images.

Three clustering algorithms were implemented and compared: K-Means, Agglomerative Clustering, and DBSCAN. The dataset was preprocessed using feature extraction and PCA for dimensionality reduction. Finally, t-SNE was used to visualize the clustering results in 2D space.

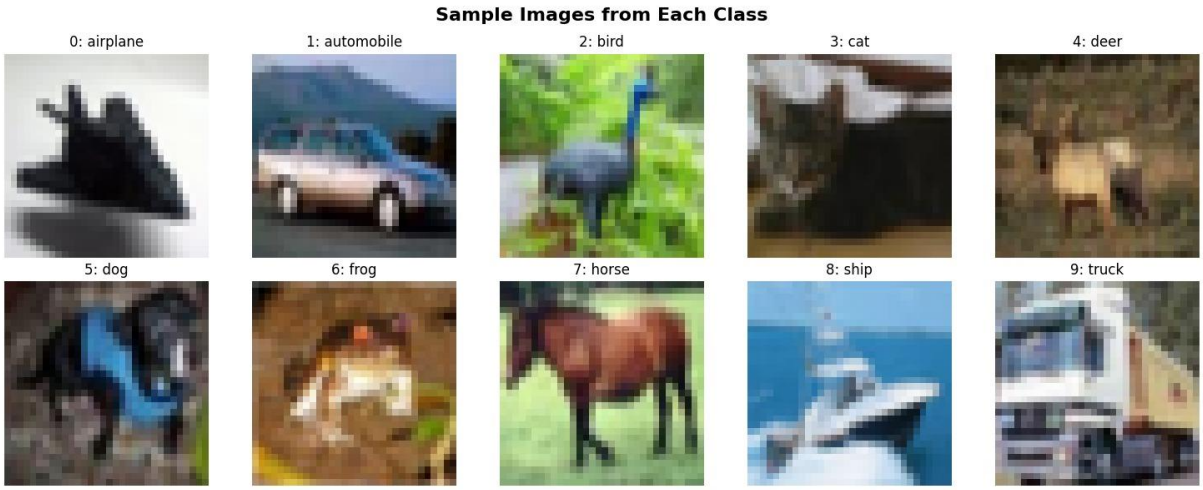
2. Dataset Overview

2.1 CIFAR-10 Dataset Description

The CIFAR-10 dataset consists of 60,000 color images of size 32×32 pixels, divided into 10 classes. Each class contains 6,000 images. The dataset is split into a training set of 50,000 images and a test set of 10,000 images. For this homework, only the training set was used for clustering analysis.

2.2 Class Names

The CIFAR-10 dataset contains images from the following 10 classes:



2.3 Data Loading

The dataset was loaded using the Keras API from TensorFlow. Sample images from each class were displayed to verify the data was loaded correctly and to understand the visual characteristics of each class.

3. Feature Extraction and Dimensionality Reduction

3.1 Feature Extraction Process

To prepare the images for clustering algorithms, the following preprocessing steps were performed:

1. Each 32×32×3 image was flattened into a 1D vector of 3,072 dimensions (32 × 32 × 3 = 3,072).

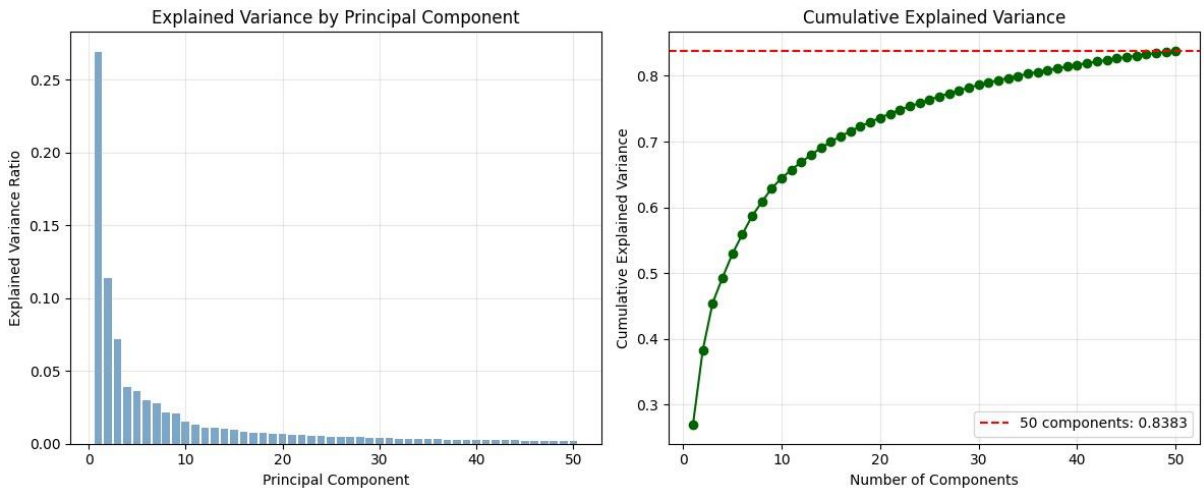
2. The flattened features were scaled using StandardScaler to normalize the data (mean = 0, standard deviation = 1).
3. Principal Component Analysis (PCA) was applied to reduce the dimensionality from 3,072 to 50 components.

3.2 PCA Results

PCA was used to reduce the high-dimensional image data to a more manageable size while retaining most of the variance in the data.

Metric	Value
Original Dimensions	3,072
Reduced Dimensions	50
Dimension Reduction	98.37%
Total Explained Variance	83.83%
First Component Variance	26.92%
Second Component Variance	11.37%

The results show that with only 50 principal components (1.63% of the original features), we retained 83.83% of the total variance in the data. This indicates that the high-dimensional image data contains significant redundancy that can be effectively compressed.



3.3 Stratified Sampling

To reduce computational time and avoid memory issues, stratified random sampling was applied to the dataset. This approach was recommended in the assignment instructions to maintain class proportions while working with a smaller subset of the data.

Parameter	Value
Original Dataset Size	50,000 samples
Sampled Dataset Size	3,000 samples
Sampling Percentage	6%
Samples per Class	300
Class Distribution	10% per class (balanced)

The stratified sampling ensured that each class was equally represented in the sampled dataset, maintaining the statistical properties of the original data. This approach reduced computation time by approximately 17 times while preserving the ability to draw valid conclusions from the analysis.

4. Clustering Methods

4.1 K-Means Clustering

4.1.1 Algorithm Description

K-Means is a centroid-based clustering algorithm that partitions the data into k clusters by minimizing the within-cluster sum of squares. It iteratively assigns each data point to the nearest centroid and updates the centroids based on the assigned points.

4.1.2 Configuration

- Number of clusters (k): 10
- Algorithm: Elkan (optimized variant)
- Maximum iterations: 300
- Number of initializations: 10
- Random state: 42 (for reproducibility)

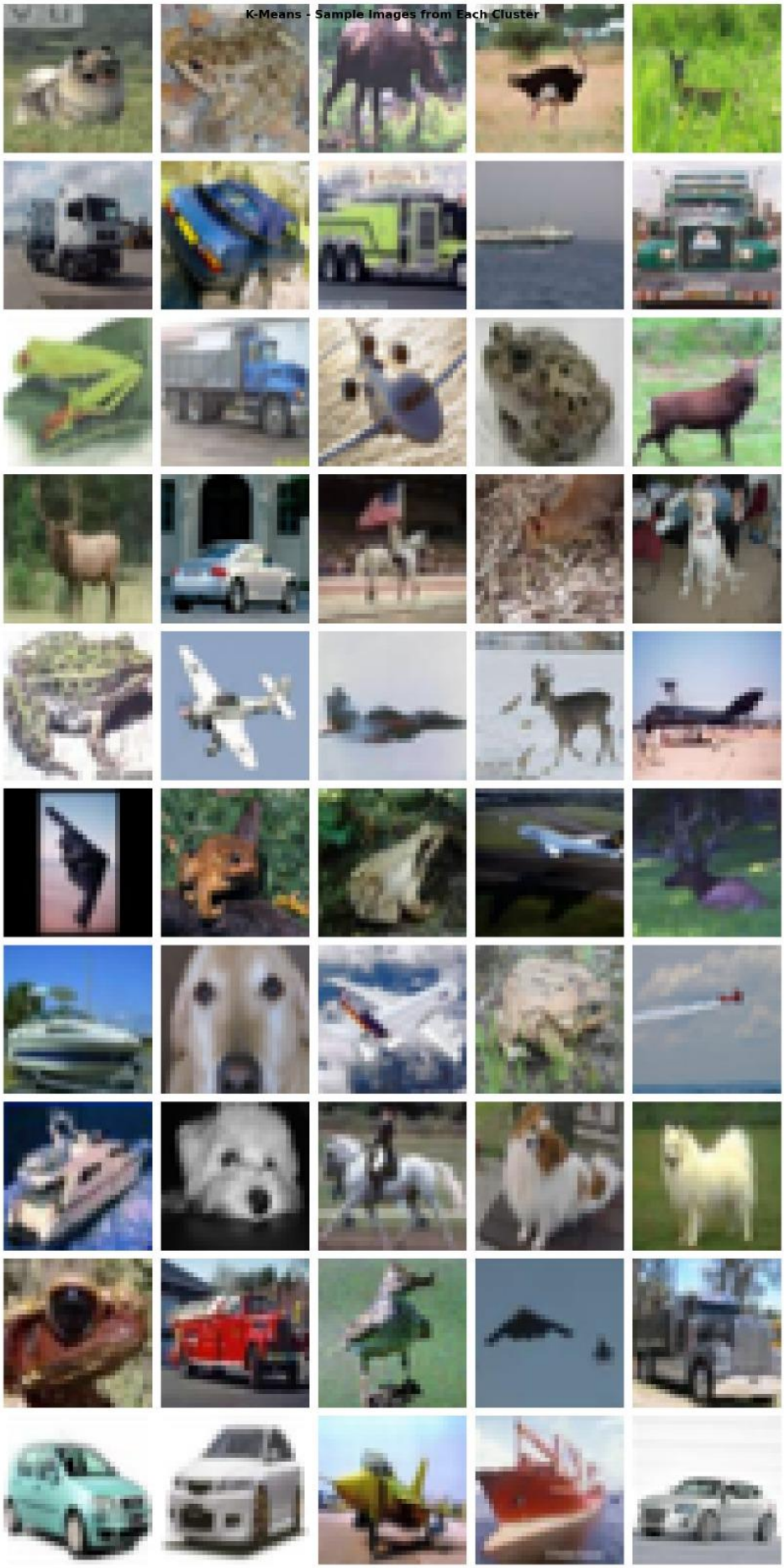
4.1.3 Results

K-Means successfully created 10 clusters with a total inertia (within-cluster sum of squares) of 4,593,403.39. The cluster sizes were relatively balanced, as shown in the table below:

Cluster ID	Number of Images	Percentage
0	324	10.8%
1	333	11.1%
2	287	9.6%
3	240	8.0%
4	189	6.3%
5	292	9.7%
6	296	9.9%
7	268	8.9%
8	518	17.3%
9	253	8.4%

4.1.4 Observations

- K-Means created relatively balanced clusters with sizes ranging from 6.3% to 17.3%.
- The algorithm executed quickly due to the optimized Elkan variant.
- Each cluster contained a mix of different object types, as the algorithm does not have semantic understanding of the images.
- The algorithm works well for spherical clusters but may struggle with complex or irregular cluster shapes.


K-Means - Sample Images from Each Cluster

4.2 Agglomerative Clustering

4.2.1 Algorithm Description

Agglomerative Clustering is a hierarchical clustering method that builds a tree of clusters using a bottom-up approach. It starts with each data point as a separate cluster and iteratively merges the closest pairs of clusters until the desired number of clusters is reached.

4.2.2 Configuration

- Number of clusters: 10
- Linkage method: Ward (minimizes within-cluster variance)
- Affinity: Euclidean distance

4.2.3 Results

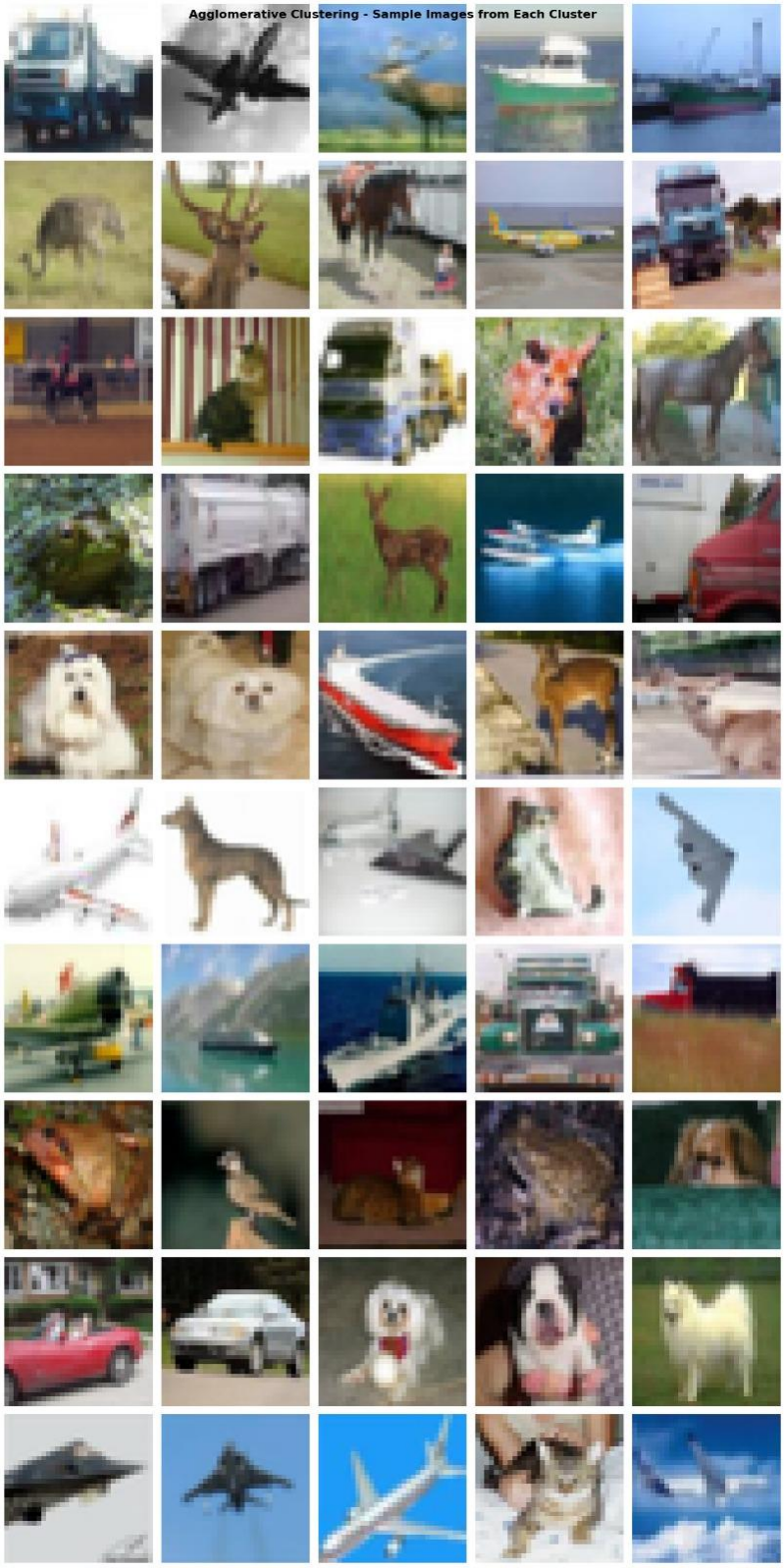
Agglomerative Clustering produced 10 clusters with more varied sizes compared to K-Means:

Cluster ID	Number of Images	Percentage
0	391	13.0%
1	352	11.7%
2	523	17.4%
3	402	13.4%
4	360	12.0%
5	114	3.8%
6	248	8.3%
7	192	6.4%
8	327	10.9%
9	91	3.0%

4.2.4 Observations

- Cluster sizes were more varied, ranging from 3.0% to 17.4%.
- The algorithm can capture more complex, non-spherical cluster shapes.

- The hierarchical structure provides additional information about relationships between data points.
- Computation time was slower than K-Means but still acceptable for the sampled dataset.



4.3 DBSCAN

4.3.1 Algorithm Description

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) is a density-based clustering algorithm that groups together points that are closely packed together and marks points in low-density regions as outliers (noise).

4.3.2 Configuration

- Epsilon (eps): 5.5
- Minimum samples: 15
- Algorithm: Ball tree (optimized for high dimensions)

4.3.3 Results

DBSCAN failed to identify any clusters in the PCA-reduced feature space. All 3,000 samples were classified as noise points.

4.3.4 Analysis

The failure of DBSCAN to identify any clusters can be attributed to several factors:

- The curse of dimensionality: Even with PCA reduction to 50 dimensions, Euclidean distances may not effectively capture density in high-dimensional spaces.
- CIFAR-10 images are highly diverse within classes, resulting in diffuse rather than dense clusters.
- The parameters (eps and min_samples) are highly sensitive and difficult to tune for high-dimensional data.
- PCA transformation may have dispersed the density patterns that DBSCAN relies on.

4.4 Comparison of Clustering Algorithms

Algorithm	Clusters Found	Cluster Balance	Speed	Suitability
K-Means	10	Good (6.3% - 17.3%)	Fast	Good for initial exploration
Agglomerative	10	Moderate (3.0% - 17.4%)	Moderate	Better for complex shapes
DBSCAN	0	N/A	Fast	Not suitable for this data

5. Visualization and Analysis

5.1 t-SNE Dimensionality Reduction

t-SNE (t-Distributed Stochastic Neighbor Embedding) was applied to reduce the 50-dimensional PCA features to 2 dimensions for visualization purposes. t-SNE is particularly effective for visualizing high-dimensional data because it preserves local structure and reveals clusters in the data.

5.1.1 Configuration

- Input dimensions: 50 (PCA-reduced)
- Output dimensions: 2
- Perplexity: 30
- Maximum iterations: 500
- Learning rate: 200

5.1.2 Results

t-SNE successfully reduced the 50-dimensional data to 2 dimensions with a final KL divergence of 2.174346, indicating good convergence. The mean sigma was 12.415, representing the effective local neighborhood size.

5.2 Ground Truth Visualization

When the t-SNE results were colored by the true class labels, several observations were made:

- The plot shows natural groupings of the 10 CIFAR-10 classes.
- Some classes are well-separated, particularly vehicles (ships, airplanes, trucks, automobiles) versus animals.
- Significant overlap exists between similar animal classes (cats, dogs, deer, horses).
- This inherent overlap demonstrates the difficulty of unsupervised image clustering when classes have similar visual features.

5.3 K-Means Visualization

The t-SNE plot colored by K-Means cluster assignments showed:

- Relatively compact and spherical clusters.
- Cluster boundaries that are approximately linear.
- Some clusters capture semantically mixed content due to the lack of semantic understanding.
- The algorithm performs best when natural clusters are roughly spherical and well-separated.

5.4 Agglomerative Clustering Visualization

The t-SNE plot colored by Agglomerative Clustering assignments revealed:

- More complex, non-spherical cluster shapes.
- Better adaptation to the natural boundaries visible in the ground truth.
- Hierarchical relationships between data points.
- Ability to capture varying density in different regions.

5.5 DBSCAN Visualization

Since DBSCAN classified all points as noise, the t-SNE plot showed all points in a single category. This visualization confirmed the algorithm's inability to identify dense regions in the PCA-reduced feature space.

5.6 Class Separability Analysis

By comparing the clustering results with ground truth labels, we identified which classes are easier or harder to cluster:

Well-Separated Classes:

- Ships: Distinct shape and color, often with water background.
- Airplanes: Distinctive wing shapes and often sky backgrounds.
- Trucks and Automobiles: Clear vehicle features and rectangular shapes.

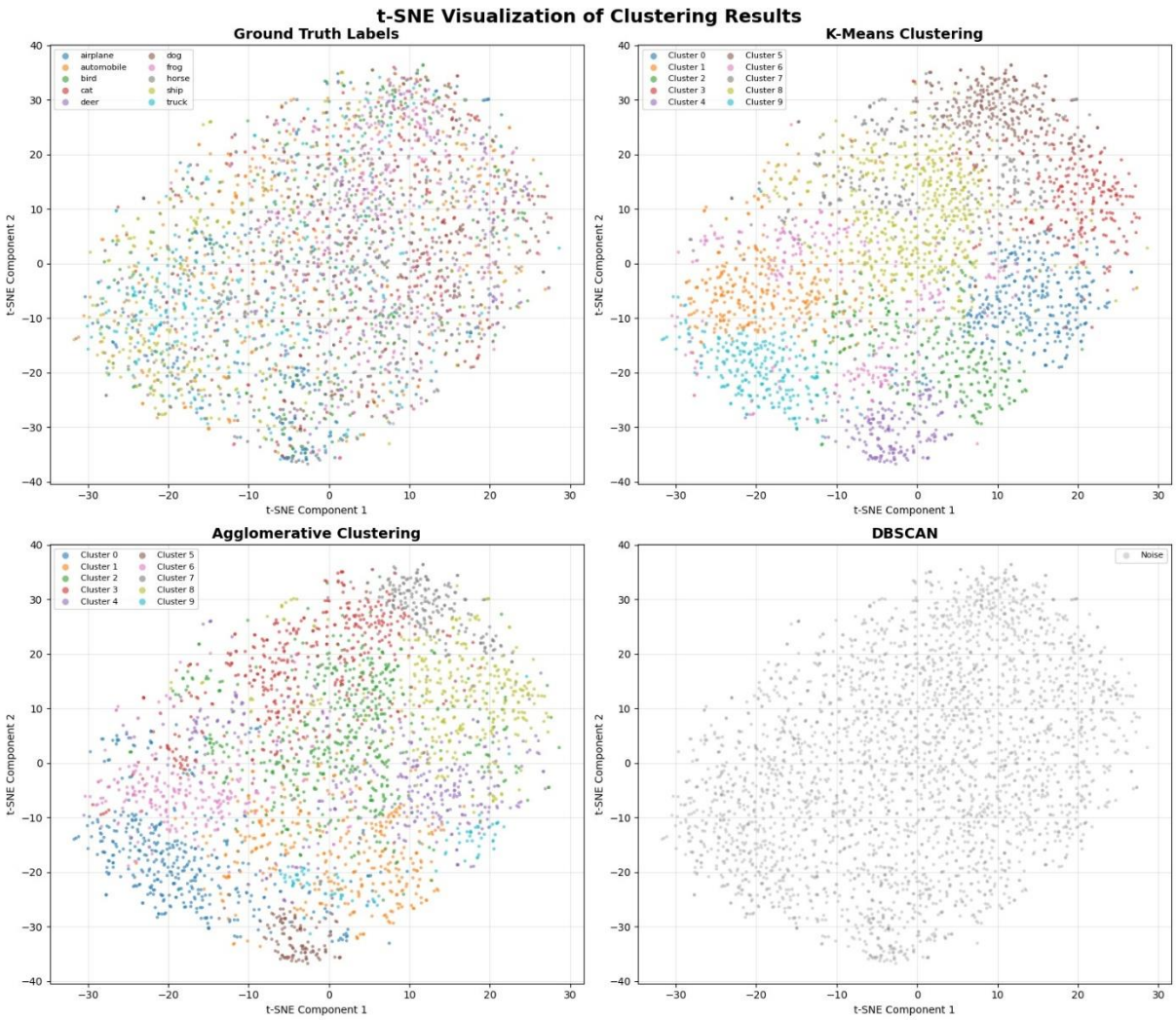
Overlapping Classes:

- Cats, Dogs, Deer, and Horses: Similar animal body structures, comparable poses, and shared textures.
- Birds and Frogs: Small animals with high intra-class variance and varied appearances.

6. Discussion and Conclusions

6.1 Key Findings

1. PCA successfully reduced dimensionality by 98.37% while retaining 83.83% of the variance, demonstrating significant redundancy in raw pixel features.
2. K-Means provided the most balanced and interpretable clustering results, making it suitable for initial exploratory analysis.



3. Agglomerative Clustering offered more flexibility in capturing complex cluster shapes, though with less balanced cluster sizes.
4. DBSCAN failed to identify any clusters, highlighting the challenges of density-based methods in high-dimensional spaces.
5. Vehicle classes showed better separability than animal classes in the feature space.
6. Stratified sampling with only 6% of the data provided sufficient statistical validity while achieving 17x speedup.

6.2 Algorithm Performance Summary

K-Means:

Strengths: Fast execution, balanced clusters, easy to implement and tune. Weaknesses: Assumes spherical clusters, sensitive to initialization, requires predefined number of clusters. Best for: Initial exploration and balanced partitioning.

Agglomerative Clustering:

Strengths: Captures complex shapes, provides hierarchical structure, adapts to density variations. Weaknesses: Slower than K-Means, memory intensive, produces less balanced clusters. Best for: Complex cluster shapes and hierarchical analysis.

DBSCAN:

Strengths: Finds arbitrary shapes, identifies outliers, no predefined k needed. Weaknesses: Very parameter-sensitive, poor performance in high dimensions, struggles with varying density. Best for: Low-dimensional data with clear density patterns.

6.3 Limitations

- Low resolution (32×32 pixels) limits the detail available for distinguishing similar classes.
- Raw pixel values and PCA may not capture semantic visual features effectively.
- Unsupervised clustering algorithms lack semantic understanding of object categories.
- High dimensionality affects distance metrics, even after PCA reduction.
- DBSCAN parameter tuning is challenging for high-dimensional image data.

6.4 Future Work Recommendations

- Use deep learning features from pre-trained CNNs (e.g., ResNet, VGG) instead of raw pixels.
- Explore alternative distance metrics such as cosine similarity or learned distance functions.
- Implement ensemble clustering methods that combine multiple algorithms.
- Try alternative dimensionality reduction techniques like UMAP or autoencoders.
- Investigate semi-supervised approaches that incorporate limited labels.
- Use silhouette score or elbow method to determine optimal number of clusters.

6.5 Conclusion

This project successfully demonstrated the application of three clustering algorithms to the CIFAR-10 image dataset. While no algorithm perfectly separated all 10 classes in an unsupervised manner, the analysis provided valuable insights into the nature of image data and the strengths and limitations of different clustering approaches. K-Means emerged as the most practical choice for this task, balancing computational efficiency with reasonable clustering quality. The hierarchical structure of Agglomerative Clustering offered additional insights, while DBSCAN's failure highlighted the challenges of density-based methods in high-dimensional feature spaces. The visualization using t-SNE confirmed that certain classes are inherently difficult to separate based solely on visual features, which has important implications for both unsupervised and supervised learning tasks.